Attention and LLMs

Unit 7 · reference sheet · AI/ML in 35 topics

Attention

$$\text{Attn}(Q,K,V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right)V$$

Q asks, K advertises, V delivers. Cost $O(n^2 d)$.
Scale by $\sqrt{d_k}$: score std grows like $\sqrt{d_k}$; unscaled softmax saturates.
Causal mask: set $j > i$ to $-\infty$ before softmax.

Block

$x \leftarrow x + \text{MHA}(\text{LN}(x))$; $x \leftarrow x + \text{FFN}(\text{LN}(x))$
Params/block at d: $4d^2$ (attention) $+8d^2$ (FFN) $= 12d^2$
Position must be injected; attention alone is permutation-invariant.

Pretraining

$\text{NLL} = -\sum_t \log p_\theta(t_t \mid t_{<t})$; perplexity $= e^{\text{NLL}}$
GPT causal (generation) vs BERT masked 15% (encoding)
loss 3.0 nats ⇒ perplexity 20.

Alignment

SFT → reward model (Bradley–Terry) → PPO $+\,\beta\,$KL
DPO: same preference optimum, no reward model, no RL.
Sampling: $p = \text{softmax}(z/T)$; $T \downarrow$ sharpens, adds no knowledge.

Gotchas

- Attention weights ≠ explanation.
- Higher temperature ≠ creativity; it raises rare tokens.
- Perplexity ≠ instruction following.

This unit at a glance

#	Topic	You can
26	Attention	compute scaled dot-product attention by hand for 2 tokens; explain Q/K/V roles
27	The Transformer block	name every sub-layer, its purpose, and why residual + LayerNorm are required
28	Pretraining	contrast next-token (GPT) vs masked (BERT) objectives; explain what scale buys
29	Alignment & inference	explain SFT → RLHF/DPO; how context window and temperature shape output

If you see this → do that

Signal	Move
Order matters	positional encoding
Must generate	causal mask
Must classify	encoder (BERT-style)
Context too long	sparse/linear attention
Inference too slow	KV cache / distill

Numbers worth memorising

d=512, 8 heads ⇒ $d_k = 64$
Block params: $12d^2 = 3.15$M at d=512
T=0.5: top prob 0.87; T=2: 0.51
perplexity $= e^{\text{loss}}$; loss 3 ⇒ 20